

ASTRAQ ET-M

18 kg-class collaborative robotic massage and therapy platform for wellness, physiotherapy and recovery applications.



ET-M Robotic Massage System

Key Configuration

Series	ET-M CR18-18/1.0
Robot architecture	6-axis collaborative robotic therapy arm
Payload	18 kg
Reach	1,062 mm
Repeatability	±0.03 mm
Typical power	600 W
Noise	≤ 70 dB(A)
Protection rating	IP54
Programming	Direct teaching control and graphical interface
Application focus	Robotic massage, rehab assistance, wellness and sports recovery

SYSTEM POSITIONING

ET-M combines a soft medical-wellness exterior with a class collaborative robot platform. It is designed for controlled, repeatable and configurable therapy paths in professional treatment spaces under operator supervision.

CORE ADVANTAGES

Force-aware therapy Built-in joint sensing supports compliant and safe interaction.	High payload margin 18 kg payload supports robust therapy end-effectors.	Practical reach 1,062 mm working reach for table-side treatment coverage.	Easy workflow setup Direct teaching and graphical programming reduce deployment time.	Clinic-ready design Quiet, smooth and premium appearance for wellness environments.
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Technical Parameters

ROBOT ARM

Model / Series	ET-M CR18-18/1.0
Robot type	6-axis collaborative robotic therapy arm
Payload	18 kg
Reach	1,062 mm
Robot weight	About 38 kg
Degrees of freedom	6
MTBF	> 80,000 h
Power supply	48 VDC
Typical power	600 W

PERFORMANCE AND MOTION

Repeatability	±0.03 mm
Force sensing	Force x-y-z: 0.1 N; Torque x-y-z: 0.02 Nm
Cartesian stiffness	0-18,000 N/m; 0-2,500 Nm/rad
Joint working range	Axis 1 ±360°; Axis 2 ±360°; Axis 3 ±168°; Axis 4 ±360°; Axis 5 ±360°; Axis 6 ±360°
Max joint speed	Axis 1 163°/s; Axis 2 163°/s; Axis 3 135°/s; Axis 4 155°/s; Axis 5 199°/s; Axis 6 228°/s

SAFETY AND ENVIRONMENT

Safety functions	More than 21 adjustable safety features including collision detection, virtual walls and collaboration mode
Safety references	EN ISO 13849-1; EN ISO 10218-1 / PL d, Cat. 3; ISO 15066; CE, KCs and EAC marking requirements
IP rating	IP54
ISO cleanroom class	Class 5
Noise	≤ 70 dB(A)
Temperature	0°C to 50°C
Humidity	≤ 93% RH, non-condensing
Installation orientation	At any angle

Max tool-end speed	≤ 6.0 m/s
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CONTROLLER AND INTERFACE

Controller	Built-in controller option for CR18-class architecture
Operator interface	Notebook / PAD / drag interactive module
Safety devices	Handheld enable and handheld emergency stop
Communication	TCP/IP 1000Mbit, Modbus TCP, Profinet, Ethernet/IP, DeviceNet, CC-Link, CC-Link IE Field Basic
Tool I/O	2 digital outputs, 2 digital inputs, 2 analog inputs
Tool communication	RS485; alternatively two analog input pins
Tool I/O power	12 V / 24 V, 1 A rated
Base output power	24 V, 1.5 A
Teach pendant	xPad2, 10.1-inch LCD, 1920 x 1200, IP54

THERAPY APPLICATION PACKAGE

Massage workflow Configurable back, shoulder and limb massage paths for treatment-bed scenarios.	End-effector options Massage, percussion, pressure or soft-tissue therapy modules can be configured.	Human-machine safety Collaborative safety setup supports shared therapist, client and robot workspaces.	Future integration Open interfaces can support treatment UI, scheduling and therapy-data platforms.
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Note: Final specifications may be refined for the ET-M therapy end-effector, software interface and clinical workflow package.